

WDBX: A Weighted-Backtrace Memory Store for Traceable Retrieval

A technical note on durable storage, approximate vector search and inspectable ranking, with legacy and canonical episode paths distinguished.

Implemented · Reviewed 2026-09-06 · MLAI Research · WDBX Core

The described source behavior is implemented; customer outcomes and deployment readiness are not established.

Practical context

A technical note on durable storage, approximate vector search and inspectable ranking, with legacy and canonical episode paths distinguished.

Build document recall and decision-history prototypes with explicit source ownership.

Implementation and research boundary

The active Rust substrate provides durable snapshots and WAL recovery, a layered HNSW graph, and a hybrid retrieval API that exposes score components. HNSW defaults in the inspected source are $M=16$, $EF_CONSTRUCTION=40$ and $EF_SEARCH=32$.

Legacy snapshot chain verification and strict content verification are different entry points. Canonical v3 episode storage is also a distinct API. Applications must identify the path they actually use; this note does not assert that every write from every consumer passed episode admission.

Similarity and approximate candidate selection

For nonzero, equal-dimension vectors, cosine similarity is the normalized dot product below. It ranges from -1 to 1; it is not a probability. The CPU reference returns zero for empty or zero-norm inputs. HNSW supplies approximate candidates rather than an exhaustive guarantee of the globally closest records. An evaluation therefore needs an exact-search baseline for its chosen corpus and requested result count.

$$\sigma(q, v) = \frac{\sum_d q_d v_d}{\sqrt{\sum_d q_d^2} \sqrt{\sum_d v_d^2}}$$

Hybrid ranking is a separate calculation

The hybrid scorer clamps semantic similarity and caller-provided persona affinity to the unit interval, then multiplies them by temporal and causal factors. With a positive half-life, temporal weight halves each half-life of nonnegative age; a nonpositive half-life disables decay. Missing timestamps use the query time. Causal reachability is bounded and uses an undirected view of persisted edges, so a hop count should not be interpreted as a proof of causal direction.

$$S_j = \text{clamp}_{[0, 1]}(\sigma_j) \tau_j \gamma_j \text{clamp}_{[0, 1]}(\pi_j)$$

$$\tau_j = 2^{-\max(0, t_0 - t_j)/t_{1/2}}$$

A reconstructible storage experiment

Use a disposable corpus with known vector neighbors and explicit metadata. Capture the source revision, vector dimensions, insertion order, query vectors and index parameters. Compare approximate results with exact ranking, then checkpoint, reopen and repeat the same queries. Separately alter a copied block payload to distinguish predecessor-link checks from strict content-hash verification. Neither successful recovery nor matching hashes validates the factual content of a stored statement. Canonical episode commitments must be tested through the v3 episode path, not inferred from legacy snapshot success.

Evaluation before adoption

Start with a synthetic or authorized corpus. Check recovery, source identifiers, update behavior and retrieval relevance. Treat application permissions, data retention and deployment evidence as additional requirements.

Limitations

Storage integrity establishes consistency, not the truth of stored statements.

The reference cluster protocol does not establish production multi-host operation or sharding.

Legacy snapshot chain verification and strict content verification are different entry points. Canonical v3 episode storage is also a distinct API. Applications must identify the path they actually use; this note does not assert that every write from every consumer passed episode admission.

Pinned implementation sources

Durable store and writer ownership

<https://github.com/donaldfilimon/wdbx/blob/14cb1341cb454bd3f887c4e54a83f8c42775a91d/crates/abi-wdbx/src/durable.rs>

Layered HNSW index

<https://github.com/donaldfilimon/wdbx/blob/14cb1341cb454bd3f887c4e54a83f8c42775a91d/crates/abi-wdbx/src/hnsw.rs>

Hybrid retrieval and observable score factors

<https://github.com/donaldfilimon/wdbx/blob/14cb1341cb454bd3f887c4e54a83f8c42775a91d/crates/abi-wdbx/src/retrieval.rs>

Snapshot chain verification

<https://github.com/donaldfilimon/wdbx/blob/14cb1341cb454bd3f887c4e54a83f8c42775a91d/crates/abi-wdbx/src/store.rs>

Canonical episode persistence

<https://github.com/donaldfilimon/wdbx/blob/14cb1341cb454bd3f887c4e54a83f8c42775a91d/crates/abi-wdbx/src/v3/episode/store.rs>

Temporal and causal ranking functions

<https://github.com/donaldfilimon/wdbx/blob/14cb1341cb454bd3f887c4e54a83f8c42775a91d/crates/abi-wdbx/src/temporal.rs>

CPU cosine reference and vector edge cases

<https://github.com/donaldfilimon/wdbx/blob/14cb1341cb454bd3f887c4e54a83f8c42775a91d/crates/abi-compute/src/cpu.rs>

This edition supersedes the June 2026 PDF. The original remains available as historical material; its broader claims must not be treated as current implementation evidence.